

Bayes Factors for Peri-Null Hypotheses

Alexander Ly · Eric-Jan Wagenmakers

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Abstract A perennial objection against Bayes factor point-null hypothesis tests is that the point-null hypothesis is known to be false from the outset. We examine the consequences of approximating the sharp point-null hypothesis by a hazy ‘peri-null’ hypothesis instantiated as a narrow prior distribution centered on the point of interest. The peri-null Bayes factor then equals the point-null Bayes factor multiplied by a correction term which is itself a Bayes factor. For moderate sample sizes, the correction term is relatively inconsequential; however, for large sample sizes the correction term becomes influential and causes the peri-null Bayes factor to be inconsistent and approach a limit that depends on the ratio of prior ordinates evaluated at the maximum likelihood estimate. We characterize the asymptotic behavior of the peri-null Bayes factor and briefly discuss suggestions on how to construct peri-null Bayes factor hypothesis tests that are also consistent.

Keywords Consistency · Peri-null correction factor · Asymptotic sampling distribution

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A. Ly
University of Amsterdam
Centrum Wiskunde & Informatica
E-mail: a.ly@uva.nl

E.-J. Wagenmakers
University of Amsterdam
E-mail: ejwagenmakers@gmail.com

vagueness leads nowhere.

Jeffreys, 1937

1 Introduction

In the Bayesian paradigm, the support that data $y^n := (y_1, \dots, y_n)$ provide for an alternative hypothesis $\mathcal{H}_1$ versus a point-null hypothesis $\mathcal{H}_0$ is given by the Bayes factor $\text{BF}_{10}(y^n)$:

$$\underbrace{\frac{p(y^n | \mathcal{H}_1)}{p(y^n | \mathcal{H}_0)}}_{\text{BF}_{10}(y^n)} = \frac{\overbrace{P(\mathcal{H}_1 | y^n)}^{\text{Posterior model odds}}}{\overbrace{P(\mathcal{H}_0 | y^n)}^{\text{prior model odds}}} \bigg/ \frac{\overbrace{P(\mathcal{H}_1)}^{\text{prior model odds}}}{\overbrace{P(\mathcal{H}_0)}^{\text{prior model odds}}} \quad (1)$$

$$= \frac{\int_{\Theta_1} f(y^n | \theta_1) \pi(\theta_1 | \mathcal{H}_1) d\theta_1}{\int_{\Theta_0} f(y^n | \theta_0) \pi(\theta_0 | \mathcal{H}_0) d\theta_0}, \quad (2)$$

where the first line indicates that the Bayes factor quantifies the change from prior to posterior model odds (Wrinch and Jeffreys, 1921), and the second line indicates that this change is given by a ratio of marginal likelihoods, that is, a comparison of prior predictive performance obtained by integrating the parameters θ_j out of the j th model's likelihood $f(y^n | \theta_j)$ at the observations y^n with respect to the prior density $\pi(\theta_j | \mathcal{H}_j)$ (Jeffreys, 1935, 1939; Kass and Raftery, 1995). Although the general framework applies to the comparison of any two models (as long as the models make probabilistic predictions; Dawid, 1984; Shafer and Vovk, 2019), the procedure developed by Harold Jeffreys in the late 1930s was explicitly designed as an improvement on p -value null-hypothesis significance testing.

In the prototypical scenario, a null hypothesis $\mathcal{H}_0$ has p_0 free parameters, whereas an alternative hypothesis $\mathcal{H}_1$ has $p = p_0 + 1$ free parameters; the additional free parameter in $\mathcal{H}_1$ is the one that is *test-relevant*. For instance, in Jeffreys's t -test the test-relevant parameter $\delta = \mu/\sigma$ represents standardized effect size; after assigning prior distributions to the models' parameters we may compute the Bayes factor in favor of $\mathcal{H}_0 : \delta = 0$ with free parameter $\theta_0 = \sigma \in (0, \infty)$ against $\mathcal{H}_1 : \theta_1 = (\delta, \sigma) \in \mathbb{R} \times (0, \infty)$ where δ is unrestricted and where σ denotes the common nuisance parameter. When $\text{BF}_{01}(y^n) = 1/\text{BF}_{10}(y^n)$ is larger than 1, the data provide evidence that the 'general law' $\mathcal{H}_0$ can be retained; when $\text{BF}_{01}(y^n)$ is smaller than 1, the data provide evidence for $\mathcal{H}_1$, the model that relaxes the general law. The larger the deviation from 1, the stronger the evidence. Importantly, in Jeffreys's framework the test-relevant parameter is fixed under $\mathcal{H}_0$ and free to vary under $\mathcal{H}_1$. The hypothesis $\mathcal{H}_0$ is generally known as a 'point-null' hypothesis.

A perennial objection against point-null hypothesis testing—whether Bayesian or frequentist—is that in most practical applications, the point-null is never

true exactly (e.g., [Bakan, 1966](#); [Berkson, 1938](#); [Edwards et al., 1963](#); [Jones and Tukey, 2000](#); [Kruschke and Liddell, 2018](#); see also [Laplace, 1774/1986](#), p. 375). If this argument is accepted and $\mathcal{H}_0$ is deemed to be false from the outset, then the test merely assesses whether or not the sample size was sufficiently large to detect the non-zero effect. This objection was forcefully made by Tukey:

“Statisticians classically asked the wrong question—and were willing to answer with a lie, one that was often a downright lie. They asked “Are the effects of A and B different?” and they were willing to answer “no.”

All we know about the world teaches us that the effects of A and B are always different—in some decimal place—for any A and B. Thus asking “Are the effects different?” is foolish. ([Tukey, 1991](#), p. 100)

This perennial objection has been rebutted in several ways (e.g., [Jeffreys, 1937, 1961](#); [Kass and Raftery, 1995](#)); in the current work we focus on the most common rebuttal, namely that the point-null hypothesis is a mathematically convenient approximation to a more realistic ‘peri-null’ ([Tukey, 1995](#)) hypothesis $\mathcal{H}_{\tilde{0}}$ that assigns the test-relevant parameter a distribution tightly concentrated around the value specified by the point-null hypothesis (e.g., [Good, 1967](#), p. 416; [Berger and Delampady, 1987](#); [Cornfield, 1966, 1969](#); [Dickey, 1976](#); [Edwards et al., 1963](#); [George and McCulloch, 1993](#); [Jeffreys, 1935, 1936](#); [Gallistel, 2009](#); [Rousseau, 2007](#); [Rouder et al., 2009](#)). For instance, in the case of the t -test the peri-null $\mathcal{H}_{\tilde{0}}$ could specify $\delta \sim \pi(\delta | \mathcal{H}_{\tilde{0}}) = \mathcal{N}(0, \kappa_0^2)$, where the width κ_0 is set to a small value.

Previous work has suggested that the approximation of a point-null hypothesis by an interval is reasonable when the width of that interval is half a standard error in width ([Berger and Delampady, 1987](#); [Rousseau, 2007](#)) or one standard error in width ([Jeffreys, 1935](#)). Here we explore the consequences of replacing the point-null hypothesis $\mathcal{H}_0$ by a peri-null hypothesis $\mathcal{H}_{\tilde{0}}$ from a different angle. We alter only the specification of the null-hypothesis $\mathcal{H}_0$, which means that the alternative hypothesis $\mathcal{H}_1$ now overlaps with $\mathcal{H}_{\tilde{0}}$.

Below we show, first, that the effect on the Bayes factor of replacing $\mathcal{H}_0$ with $\mathcal{H}_{\tilde{0}}$ is given by another Bayes factor, namely that between $\mathcal{H}_0$ and $\mathcal{H}_{\tilde{0}}$ (cf. [Morey and Rouder, 2011](#), p. 411). This ‘peri-null correction factor’ is usually near 1, unless sample size grows large. For large sample sizes, we demonstrate that the Bayes factor for the peri-null $\mathcal{H}_{\tilde{0}}$ versus the alternative $\mathcal{H}_1$ is bounded by the ratio of the prior ordinates evaluated at the maximum likelihood estimate. This proves earlier statements from [Morey and Rouder \(2011, pp. 411–412\)](#) and confirms suggestions in [Jeffreys \(1961, p. 367\)](#) and [Jeffreys \(1973, p. 39, Eq. 2\)](#). In other words, the Bayes factor for the peri-null hypothesis is inconsistent.

Note that there exist several Bayes factor methods that have replaced point-null hypotheses with either peri-null hypotheses (e.g., Stochastic Search